MILITARY SPECIFICATION

MOLDING COMPOUNDS (FOR PLASTIC INHIBITORS)

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers molding compounds for the manufacture of inhibitors. (See 6.1)

1.2 Classification. The molding compounds covered by this specification shall be of the following classes, as specified. (See 6.2.)

Class 1 Ethyl cellulose
Class 2 Cellulose acetate

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issues in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein. In the event of conflict between this specification and any document referenced herein, requirements of this specification shall apply.

SPECIFICATIONS

Federal

*	PPP-B-1055	Barrier Material, Waterproofed Flexible
*	PPP-B-1806	Barrel and Kegs, Wood Slack

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Military

MIL-I-3166	Inhibitor, Web and End (Plastic)
MIL-A-3167	Adhesive (For Plastic Inhibitors)
MIL-P-20306	Powder, Ignition Cartridge, (M9), 60MM, 81MM Mortar

STANDARDS

Federal

FED-STD-406	Plastics: Method of Testing

Military

* MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

* Naval Sea Systems Command (Code Ident 10001)

446714	Long Web Inhibitors
446715	Short Web Inhibitors
451264	Front End Inhibitors
451265	Rear End Inhibitors

* (Copies of specification, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Lot.

3.1.1 Pilot lot. Prior to production under contract the manufacturer shall be required to produce a pilot lot of molding compound made at the plant in which he is to fill his contract, from the same material, and by the same process(es) that he proposes to follow in executing the contract.

3.1.2 Production lot. Unless otherwise specified, a production lot shall consist of the same class of molding compound prepared as a unit of manufacture for shipment at one time.

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3.2 Material. The material shall consist of suitable ethyl cellulose or cellulose acetate, plasticizer(s) and stabilizer(s) with or without the addition of dyes so compounded as to be suitable for extrusion or injection molding.

* 3.2.1 Plasticizer. The contractor shall state the chemical name of any plasticizer(s) used in the molding compounds and sample(s) shall be submitted to the procuring activity when requested.

3.2.2 Cellulose modification. The manufacturer shall state the average percent of ethylation, in the case of class 1, and acetylation in the case of class 2, of the cellulose used in the molding compound and shall describe the method of analysis used in making the determination.

* 3.3 Color. Unless otherwise specified the molding compounds shall be furnished uncolored. When colored molding compounds are specified, the color shall match the sample furnished by the procuring activity. (See 6.2 and 6.3)

* 3.4 Dimensional stability. Extruded web inhibitors and injection molded end inhibitors, made from the compounds specified herein, shall meet the dimensional stability requirements specified in 3.4.1 and 3.4.2. (See 4.3.1.2.)

3.4.1 Web inhibitors. After heating a 4-3/4-inch length of the web inhibitor for 24 hours at 140°F, it shall not warp in excess of 0.050 inch, and any change in length and width shall be less than 2 percent.

3.4.2 End inhibitors. After heating for 24 hours at 140°F, end inhibitors shall not warp in excess of 0.030 inch, and any change in length and width shall be less than 2 percent.

* 3.5 Adhesion of inhibitors to double-base powder grain. Web inhibitors, extrusion molded in accordance with MIL-I-3166 using the class of compound specified (see 6.2) shall adhere acceptably when tested as specified in 4.3.1.3.

* 3.6 Cycling and burning tests. Inhibitors made from the specified compound shall meet such cycling and burning tests as specified in 4.3.1.4.

* 3.7 Nitroglycerine absorption. The nitroglycerine absorption of extrusion or injection molded strips (1 inch by 3 inches by 0.10 inch) shall not exceed the following at any time during 3 weeks contact with double-base powder. (See 4.3.1.5.)

	Percent (max.)
Class 1	20
Class 2	70

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3.8 Physical requirements. The compounds covered by this specification shall conform to table I.

Table I. - Physical requirements

Property	Class 1	Class 2
Flow temperature, °F. (max.)- - - - - - - - - - - -	300	300
Specific gravity, 77/77° F., (max.)- - - - - - -	1.14	1.32
Heat distortion, °F., (min.)- - - - - - - - - - -	115	115
Tensile strength, (min.), psi at 77°F. - - - - -	3,500	3,500
Flexural strength - - - - - - - - - - - - - - - - -	4,000	5,000
Impact strength (I ZOD), (min.), ft.- lb per inch of notch at 77°F - - - - - - - - - - - - - - - -	4.0	1.5
at - - 40°F - - - - - - - - - - - - - - - -	1.0	0.3
Water absorption (24 hours):		
Weight gain, (max.) percent - - - - - - - - -	1.8	3.0
Soluble matter lost, (max.) percent - - - - -	0.2	0.8
Weight lost on heating (72 hrs at 180°F.) (max.) percent - - - - - - - - - - - - - - - - - - -	1.4	5.1
Flammability, (max.), inches per minute - - - - -	1.0	1.5
Light transmission,(min.), percent - - - - - - -	50	65
Modulus of elasticity in flexure, psi by 105 - -	1.1 to 1.4	1.4 to 1.9

* 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling.

* 4.2.1 Sampling for pilot lot acceptance tests. Samples shall consist of five injection-molded standard test specimens for each of the requirements specified in table I prepared in accordance with FED-STD-406 from the pilot lot of molding compound. (See 4.3.1.6.) In addition 240 extrusion-molded web inhibitors and 5 front and 5 rear injection-molded end inhibitors (size to be specified by the procuring

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activity) and 10 specimens, 1 inch wide by 3 inches long by 0.10 inch thick, from the molding compound. The samples shall be securely packed for shipment to the Government laboratory designated by the procuring activity for inspection and tests and marked with the following information:

Samples for pilot lot acceptance tests in accordance with
MIL-M-3165
Name of manufacturer
Manufacturer's designation
Authorized by (reference letter or contract number)

* 4.2.2 Sampling for production lot acceptance tests. Unless otherwise approved, no molding compound shall be accepted as a production lot until the pilot lot has been tested and found to conform to this specification. Molding compound sufficient for injection molding three specimens each for determination of flow temperature and specific gravity shall be selected.

4.3 Tests.

4.3.1 Pilot lot tests.

4.3.1.1 Color. Color of the molding compounds covered herein shall be determined on extrusion-molded web inhibitors, by visual comparison with the color sample submitted with the contract or order.

4.3.1.2 Dimensional stability. Dimensional stability shall be determined in accordance with the method specified in MIL-I-3166.

4.3.1.3 Adhesion of web inhibitors to double-base powder grain (cruciform shape). The adhesion of each of four web inhibitors to the double-base powder grains by use of an adhesive conforming to MIL-A-3167 shall be tested.

4.3.1.4 Cycling and burning. The adhesion of the proper number of web and end inhibitors (see 4.2.1) to 30 propellant grains Mk 18 Mod 0 (cruciform-shape) by use of an adhesive conforming to MIL-A-3167 shall be tested. The inhibited grain shall be cycled for 96 hours at a temperature between -10° and 140°F (24 hours at each temperature per cycle). At the completion of the temperature cycling tests, the web and end inhibitors shall be firmly and completely bonded to the grain, with the following permissible exceptions:

a. There may be an unbonded border around each or any inhibitor extending inwardly from the outer edges a distance not exceeding 0.062 inch. In addition there may exist a total length not to exceed 2.0 inches of unbonded border extending inwardly a distance not exceeding 0.125 inch.

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b. There may be an accumulative length of not more than 1 inch of unbonded border on not more than two web inhibitors which extends inwardly from the edge of the inhibitor a distance not exceeding 0.186 inch. In addition, there may be an accumulative length of not more than 0.5 inch of unbonded border on not more than two web inhibitors which extends inwardly from the edge of the inhibitor a distance not exceeding 0.250 inch.

c. The total length of unbonded areas having an inward depth between 0.062 inch and 0.250 inch must not exceed 2 inches for each of the two inhibitors mentioned in b.

d. Any web or end inhibitor overhang shall count on the accumulative total of unbonded border area for that inhibitor.

e. There may be a maximum of four unbonded corner areas for each end inhibitor which do not extend inwardly from the inhibitor corners beyond a line originating on the straight edge of the inhibitor 0.375 inch from the corner and running at an angle of 45° from the straight edge to the outer curved edge of the inhibitor.

f. In addition to the unbonded borders described in a, b, c, d, and e, there may exist isolated unbonded areas which do not extend to the edge of any inhibitor. The total area of such isolated unbonded portions shall not exceed 1.20 square inches for each of the long web inhibitors (see Drawing 446714), nor 0.60 square inch for each short web inhibitor (see Drawing 446715) nor 0.90 square inch for each of the end inhibitors (see Drawings 451264 and 451269). The area of any isolated unbonded portion shall be determined as the product of the two greatest dimensions when measured at right angles to each other.

Any of the above exceptions may be cause for rejection unless otherwise approved. Partial burning tests shall be conducted in standard partial burning equipment, on the cruciform-shape powder grain with inhibitors intact, at temperatures of -10° and 140°F, with controls set so as to interrupt the burning at approximately 50 percent of the normal burning time at each test temperature. The inhibitors on these partially burned grains shall be inspected for any abnormalities which may be cause for rejection.

* 4.3.1.5 Nitroglycerine absorption tests. Two test specimens, 1 inch by 3 inches by 0.10 inch, described in 4.2.1 shall be conditioned for 48 hours at 77° ± 2°F, and approximately 50 percent relative humidity and weight to the nearest 0.001 gram. This can most conveniently be accomplished in a closed vessel with the specimen placed above a saturated aqueous solution of magnesium nitrate hexahydrate. Place one specimen into a standard surveillance bottle containing approximately 120 grams of 81-mm powder conforming to MIL-P-20306 so as to completely surround the specimen, and cover with a watch glass. Place the other specimen into a similar bottle and cover with a watch glass (this sample will constitute a blank). Place both bottles in an oven maintained at 150° ± 2°F for 3

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weeks. Every 4 days, remove the bottles from the oven, cool to 77° ± 2°F, remove the specimens therefrom, weigh, and replace in their respective bottles. Prior to replacing the specimen in the surveillance bottle, the powder also shall be removed from the bottle and thoroughly mixed and then poured around the specimen. At the end of the specified heating period, cool the bottles to 77° ± 2°F, remove the specimens from their respective bottles and weigh. By algebraic addition correct the gain in weight of the test specimen for the change in weight of the control specimen arising from such causes as plasticizer loss. Calculate the percentage of net gain in weight since the beginning of the test. If at any period in the 3-week test the net gain in weight exceeds the amounts specified in 3.7, the lot shall be rejected.

4.3.1.6 Tests for physical requirements. The five injected-molded test specimens provided for each of the requirements of table I shall be tested in accordance with the applicable test methods of FED-STD-406. The pilot lot of molding compound is acceptable when the average result of the five test specimens conform to table I.

4.3.2 Production lot tests. Tests shall be conducted for flow temperature and specific gravity on three specimens from each production lot in accordance with FED-STD-406. The production lot shall be acceptable if the average of the tests on three specimens, for each requirement property, differ from the average of the similar pilot lot tests by not more than 5 percent of the pilot lot average.

4.4 Resubmission. A lot that has been rejected under the provisions of this specification may be resubmitted for inspection and testing provided the contractor (after being informed of the reasons for rejection) submits a signed statement that he has corrected the deficiencies noted. When any lot fails to conform to this specification and before proceeding with any further production, resubmission of samples for pilot lot tests in accordance with 4.3 shall be required.

5. PREPARATION FOR DELIVERY

* 5.1 Packing. Unless otherwise specified, the material shall be packed in slack barrels conforming to PPP-B-1806 which have been previously lined with a waterproof barrier material conforming to PPP-B-1055.

* 5.2 Marking. In addition to any special marking required in the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The molding compounds covered by this specification are for use in molding the web and end inhibitors covered by MIL-I-3166.

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6.2 Ordering data. Procurement documents should specify the following:

a. The title, number and date of this specification.
b. The class of molding compound. (See 1.2)
c. The government laboratory to which samples specified in 4.1.1 should be sent for testing.
d. Color of compound required. (See 3.3 and 6.3)

6.3 Color. When colored molding compounds are required, color samples will be furnished with requests for bids.

* 6.4 The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Preparing activity
Navy - OS
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DOCUMENT IDENTIFIER AND TITLE
MIL-M-3165A(OS), Molding Compounds (For Plastic Inhibitors)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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   ☐ YES ☐ NO *(If "Yes", in what way?)*

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